

Sunanda Vasanthi Tata

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EDUCATION

California State University, Fullerton	May 2026
M.S. Computer Science GPA: 3.7/4.0	
Gandhi Institute of Technology and Management, India	May 2023
B.Tech GPA: 3.68/4.0	

TECHNICAL SKILLS

Languages: Java, Python, SQL, TypeScript, JavaScript, Go, C, C++, C#, HTML5, CSS3
Backend & Distributed Systems: Spring Boot, FastAPI, Node.js, REST APIs, Microservices, Event-Driven Architecture, System Design Fundamentals, Data Structures & Algorithms
Artificial Intelligence: LLMs, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Search, Embedding Models, Scikit-learn, Hugging Face Transformers, PyTorch
Cloud & DevOps: AWS (EC2, S3, RDS), Azure, Docker, Kubernetes, Terraform, Azure DevOps, GitHub Actions, CI/CD
Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Kafka, Spark/PySpark, Query Optimization, Schema Design
Security, Observability & Quality: OAuth 2.0, JWT, RBAC, Prometheus, JUnit, Mockito, Test Automation, Agile Scrum
Frontend: React, TypeScript, Redux, Material UI, Responsive Web Design

EXPERIENCE

Pacific Life	Newport Beach, CA
Software Developer	Jun 2025 – May 2026

- Engineered loan portfolio validation services in **Java** and **Spring Boot** for **commercial and residential loan platforms**, exposing **REST APIs** consumed by **React** dashboards to surface ingestion status, compliance failures, and loan-level reporting for **300+ internal users**, eliminating direct SQL access for operations teams.
- Reduced API p95 latency by **40%** on **Kubernetes**-hosted services by introducing **Redis** caching for high-cardinality loan lookups and optimizing **Kafka** consumer throughput through partition rebalancing and batch-fetch tuning.
- Integrated **LangChain**-based anomaly detection into the loan validation pipeline to flag statistical outliers in daily ingestion files, reducing false-positive escalations to the accounting team by surfacing rule violations before downstream processing.
- Shipped **TypeScript** UI components for loan workflow approvals and exception handling into bi-weekly **CI/CD** releases via **GitHub Actions**, maintaining WCAG 2.1 AA compliance and contributing to **zero regression** incidents across four production cycles.

California State University, Fullerton	Fullerton, CA
Software Developer	Sept 2024 – May 2025

- Developed a **MERN-based** real-time monitoring platform that **supports 100+ connected devices**, designing **RESTful APIs** and implementing telemetry, alerting, and observability features to improve the visibility and reliability of the system.
- Built a containerized room discovery and booking service with full-text search and filtering, deployed on **Docker** and **Kubernetes**; optimized query patterns to reduce response time and support concurrent usage across the university campus.
- Researched and prototyped campus **AI-driven applications**, including **RAG-based assistants** and computer vision pipelines using deep learning frameworks, focusing on performance optimization and practical software integration.
- Implemented a **YOLOv8**-based computer vision system for hardware detection, optimizing image processing and inference workflows to improve object recognition reliability and real-time performance in deployment scenarios.

Uber	Bengaluru, India
Software Engineer	Sept 2023 – Jul 2024

- Reduced merchant onboarding processing time from **5 days to under 24 hours** by integrating real-time menu processing modules into core **Java** microservices using **Bazel** monorepo build tools, eliminating manual multi-step workflow verification delays.
- Revamped multi-fulfillment order tracking by consolidating delivery, grocery, and pickup workflows into a unified backend, utilizing **Redis** caching and **PostgreSQL** tuning to cut peak request latency by **20%**.
- Strengthened platform security infrastructure by standardizing internal **OAuth 2.0** and **RBAC** access models across distributed services, integrating **Prometheus** and **Jaeger** telemetry to accelerate production on-call incident triage.
- Shipped batch and real-time data ingestion pipelines for structured analytics datasets using **Python** and **PySpark** across **Apache Hudi** lakes, managing automated staging deployments via **Spinnaker** CI/CD infrastructure.

Hitachi Vantara	Bengaluru, India
Software Engineering Intern	Jan 2023 – Aug 2023

- Developed backend data transformation plugins for the **Pentaho** platform using **Java** and **SQL**, optimizing relational queries to improve reporting module execution efficiency by **15%**.
- Built automated integration and interoperability test scripts using **Python**, while assisting with containerized service deployments to **Docker** and internal **Kubernetes** staging clusters for **CI/CD** regression cycles.

PROJECTS

ChronoMind – AI Memory Assistant	Python, FastAPI, LangChain, Neo4j, Qdrant, OpenAI, PostgreSQL
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- Built a personal AI memory tool combining **RAG**, **Qdrant** vector search, and **Neo4j** graph relationships behind **FastAPI** services to reconstruct memory timelines; hybrid graph-vector retrieval improved contextual precision over flat semantic search across a multi-thousand-entry corpus.

Ride Matching and ETA Engine	Java, Spring Boot, Kafka, Redis, Kubernetes, PostgreSQL
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- Built an event-driven ride matching prototype using **Spring Boot**, **Kafka**, and **Redis Geo** to rank drivers and estimate ETAs; deployed on **Kubernetes** with geospatial **PostgreSQL** queries and Redis caching for low-latency matching.